

A pseudoscalar slow mode in pyrochlore ice dynamics: the emergent magnetic helicity at a thin cell

Lyndon Drake, with Claude Code (Anthropic Claude 5 models)

2026-09-06

Abstract

At the pyrochlore ice cell $(1, 2, 2)$, small enough to enumerate exactly, the slowest mode of the single-hexagon-flip chain is even under global arrow reversal and so invisible to the polarisation, the observable on which the chain's dynamical exponent is measured. This note identifies it. Its character under the cell's 256-element affine automorphism group is the determinant of the group element, so it is a pseudoscalar, and the natural bilinear pseudoscalar of a divergence-free field is the emergent magnetic helicity $\mathbf{q} \cdot (\mathbf{P}^* \times \mathbf{P})$. One helicity observable has overlap 0.759 with the eigenvector and the family spans 0.835 of it against a total degree-two content of 0.853, while every member of the project's earlier panel has overlap below 10^{-25} , a zero forced by symmetry rather than by amplitude. The helicity reaches $\tau_{\text{exp}} = 389.45$ attempts against an exact $1/\Delta = 389.95$, where the best earlier member reaches 263.13. The effect is geometric: measured in cubic cells at $L = 2$ to 16, the ratio of helicity to polarisation relaxation times falls monotonically to 0.4988 ± 0.0025 , the two-photon value, so no slow helicity mode survives a third thick direction. The verification is an exact spectrum by diagonalisation of the two parity blocks, symmetry analysis under arrow reversal and the derived point group, exhaustive parity checks on all 221,424 accessible states, and measured autocorrelations in cubic geometry. There is no formal certificate.

1 The chain and the cell

Pyrochlore spin ice states are the two-in-two-out configurations of Ising spins on the pyrochlore lattice (Bramwell and Gingras 2001), equivalently the Eulerian orientations of the diamond lattice whose bonds carry the spins. The smallest ice-preserving move reverses the six arrows around a hexagonal plaquette, a single spin flip creating a monopole pair instead (Castelnovo, Moessner, and Sondhi 2008). The chain studied here picks a hexagon uniformly at random and, if its arrows run head-to-tail, reverses all six with

probability $1/2$. It is reversible with the uniform stationary distribution on each flux sector, and under Henley's classical-to-quantum correspondence its generator is the ring-exchange Hamiltonian of pyrochlore ice at its Rokhsar–Kivelson point (Henley 2004; Rokhsar and Kivelson 1988; Hermele, Fisher, and Balents 2004).

Cells are blocks of (a, b, c) conventional cubic cells with periodic boundaries, holding $16abc$ spins and as many hexagons, the hexagon set derived from the coordinates rather than read off the quotient graph. The winding $\mathbf{W}$ is conserved and everything below sits at $\mathbf{W} = 0$, a sector never connected because frozen states exist in it at every size, so the object of study is its dynamically accessible component. At $(1, 2, 2)$, 64 spins and 64 hexagons give 221,628 zero-winding states in 205 components: a giant component of 221,424 states, whose transition matrix has 2,651,888 nonzeros, and 204 frozen singletons. That component is small enough to diagonalise rather than sample.

2 What an observable can see

Global arrow reversal R complements every arrow bit. It commutes with the transition matrix P , and on the giant component it has no fixed point, so it acts as a free involution with 110,712 two-element orbits. Taking one representative r_j per orbit and its partner p_j , the vectors $(e_{r_j} \pm e_{p_j})/\sqrt{2}$ span the even and odd subspaces, and $P^\pm[j, k] = P[r_j, r_k] \pm P[r_j, p_k]$ are two symmetric blocks of size 110,712 whose spectra partition the spectrum of P .

The autocorrelation of an observable O is $\rho(t) = \langle y, P^t y \rangle / \langle y, y \rangle$ with $y = O - \langle O \rangle$, a sum of decaying exponentials weighted by the squared overlaps of y on the eigenvectors, so an observable of definite parity has overlap exactly zero on every eigenvector of the opposite parity and its τ_{exp} is only a lower bound on $1/\Delta$. That an observable's autocorrelation time is not the chain's is standard (Sokal 1997). What is unusual here is that a symmetry makes the shortfall exact, so no series length and no system size closes it.

The natural observable is the coarse-grained polarisation: with $\sigma_b = 2x_b - 1$, $\mathbf{d}_b$ the bond's A-to-B displacement and $\mathbf{q}^{(k)} = (2\pi/4c_k) \mathbf{e}_k$, $P_j^{(k)} = N^{-1} \sum_b \sigma_b d_{bj} e^{i\mathbf{q}^{(k)} \cdot \mathbf{r}_b}$, whose transverse components $j \neq k$ carry the Coulomb-phase slow mode (Henley 2010; Isakov et al. 2004). Every σ_b changes sign under R , so $\mathbf{P}$ is odd, and at $(1, 2, 2)$ the three slowest eigenvectors are even. The project's response was a panel of both parities at every

size, the slowest member taken as the headline and reported as a lower bound. Its three even members were the flippability structure factor $F(\mathbf{q})$, the total flippable count, and $|\mathbf{P}_\perp|^2$. None of those sees the slow mode either, for a second reason beyond parity.

3 The exact spectrum, and a pseudoscalar character

Lanczos on each parity block, $k = 14$, with the uniform vector deflated in the even block, took 930 and 973 matrix-vector products. Merged, the three slowest modes of the chain are even and the fourth is odd. Per attempt $\Delta_{\text{even}} = 2.5644358 \times 10^{-3}$ and $\Delta_{\text{odd}} = 4.1225073 \times 10^{-3}$, a ratio of 0.622057, so $1/\Delta = 389.949$ attempts against the 242.571 the polarisation can reach. The slowest even eigenvector, written v below, is simple, separated from the second even mode by 1.229×10^{-3} in λ , so every reading of it is a property of the chain. The slowest odd eigenvalue is fourfold degenerate, which is why the odd readings here are multiplet invariants. Against an independently banked earlier spectrum the maximum difference over the top ten eigenvalues is 5.44×10^{-15} , all ten parities agree, and the lifted eigenvectors satisfy $\|Pv - \lambda v\| < 10^{-14}$.

The cell's affine automorphisms are derived from the coordinates rather than assumed: maps $p \mapsto Mp + t$ modulo the box, with M a signed axis permutation, are accepted only if they permute the site set, carry every bond to a bond, and carry the derived hexagon set to itself, and where the map exchanges the A and B sublattices the arrow bit complements. At $(1, 1, 1)$ the construction returns 192 automorphisms, the number of general positions of $Fd\bar{3}m$ in one conventional cubic cell, asserted as a test, and at $(1, 2, 2)$ it returns 256, all acting on the giant component and each verified to commute with P . The character $\chi(g) = \langle v, gv \rangle$ is ± 1 on every one of them, and

$$\max_g |\chi(g) - \det M_g| = 2.66 \times 10^{-14}$$

over 128 proper and 128 improper operations. So v transforms in the determinant representation, invariant under every proper operation and sign-reversing under every improper one, with a stabiliser of exactly 128 elements. Under the two transverse translations its character is 1.000000000000, so it sits at zero momentum, while the trace characters 0, 0 and -4 of the fourfold odd multiplet force those four states to the transverse zone-

boundary momenta. An even, zero-momentum, pseudoscalar function of a divergence-free field has a short list of candidates, and the first of them is the helicity.

4 The emergent magnetic helicity

For a divergence-free field the natural bilinear pseudoscalar is $\mathbf{q} \cdot (\mathbf{P}^* \times \mathbf{P})$. With $\mathbf{q}$ along axis k and (i, j, k) cyclic, the code computes

$$h_k(\mathbf{q}) = \text{Im}(P_i(\mathbf{q})^* P_j(\mathbf{q})) = \text{Re } P_i \cdot \text{Im } P_j - \text{Im } P_i \cdot \text{Re } P_j,$$

three multiplies and a subtraction on polarisation components the sampler already records, with P_i and P_j transverse to $\mathbf{q}$ by construction. The quantity is even under arrow reversal, being a product of two odd factors, and changes sign under every improper lattice operation because the cross product does. In the continuum it is the magnetic helicity $\int \mathbf{A} \cdot \mathbf{B}$ of the emergent gauge field. Its parity is measured rather than inferred: over all 221,424 states every helicity column satisfies $h(Rs) = h(s)$ to a relative deviation below 10^{-12} , and every non-vanishing polarisation column satisfies $P(Rs) = -P(s)$. Of 259 named observables whose overlap $\langle v, O \rangle^2 / (\|v\|^2 \|O\|^2)$ with v was computed, the eight largest are all helicities.

Table 1: Overlaps of the slowest even eigenvector with the helicity family, with the second even mode as a control.

observable	overlap with v	with the 2nd even mode
$h_y + h_z$ at $q_{\min}$	0.75859	2.4×10^{-26}
h_y, h_z at $q_{\min}$	0.43300	1.8×10^{-22}
$h_x + h_y + h_z$ at $q_{\min}$	0.42149	1.3×10^{-26}
$h_y + h_z$ at $2q_{\min}$	0.27991	9.8×10^{-27}
$h_y - h_z$ at $q_{\min}$	2.3×10^{-27}	4.2×10^{-22}
h_x at $q_{\min}$ (thin axis)	0.03464	1.6×10^{-27}

The span of the eleven independent helicity columns captures 0.83456 of v . A least-squares projection onto spin monomials $\sigma_S = \prod_{b \in S} \sigma_b$, done as a genuine Gram solve

because the monomials are not orthogonal once restricted to an ice sector, puts the total degree-two content of v at 0.853223, identical at rank cutoffs 10^{-8} , 10^{-10} and 10^{-12} . The helicity family therefore accounts for 98 per cent of everything of degree two in the mode, and the best non-helicity overlap is 0.066707. Two controls close the identification. The $y \leftrightarrow z$ swap has determinant -1 , so it must send v to $-v$, and geometrically it exchanges h_y with h_z and flips the helicity's sign, so $h_y + h_z$ is odd under it and matches, while $h_y - h_z$ is even and must be orthogonal, as the measured 2.3×10^{-27} says it is. And the matrix $C_{ij} = \langle v, \sigma_i \sigma_j \rangle$ describing the degree-two content has effective rank 18.35 with leading eigenvalues in $\pm$ pairs at ± 0.002933 , the signature of an antisymmetric pairing rather than of the square of a single bond-linear form, which would give a rank-one C of one sign.

5 The blind spot, in relaxation times

Autocorrelations were computed exactly by 4,000 sparse matrix-vector products, so the numbers below carry no statistical error.

Table 2: Exact relaxation times at $(1, 2, 2)$, against $1/\Delta = 389.95$ attempts and $\tau_{\text{exp}}(\lambda_2) = 389.45$.

observable	parity	τ_{int} (att)	τ_{exp} (att)	couples to
$P_{\perp}(q_{\min})$	odd	181.73	242.07	slowest odd mode
$F(q_{\min})$	even	49.78	—	nothing in the top 28
$n_{\text{flippable}}$	even	51.69	176.85	a fast even mode
$ P_{\perp}(q_{\min}) ^2$	even	105.77–124.64	263.13	2nd even mode
$h_y + h_z$ at $q_{\min}$	even	309.02	389.45	the slowest mode

The helicity's τ_{exp} is the chain's true relaxation time to the last digit, where the earlier

panel tops out at 263.13, short of the gap by a factor 1.61. Overlaps of the earlier panel's even members with v are 3.4×10^{-29} and below for the four flippability channels, 7.0×10^{-30} for the flippable count, and 8.9×10^{-27} and below for the four squared-polarisation channels, with last digits that move between runs. Those are symmetry-enforced zeros rather than small numbers, because the observables are true scalars under improper operations and v is a pseudoscalar, which is stronger than suppression by powers of $q_{\min}^2$. The stochastic estimator sees the mode too, which matters because cubic cells can only be sampled: with 512 replicas and 20,000 samples at 39 attempts it returns $\tau_{\text{exp}}(h_y + h_z) = 378.08 \pm 1.92$ attempts against 377.55 for the identical fit applied to the exact $\rho(t)$.

6 Cubic geometry

Exact diagonalisation cannot reach cubic cells: cubic $L = 2$ has of the order of 2×10^{11} states at $\mathbf{W} = 0$. The question went to the Monte Carlo instead, with the helicity added to the production panel as four even families and left off by default so that the earlier panel reproduces bit for bit. With the same seeds, sampling intervals, burn-ins, replica counts and series lengths, $\tau_{\text{exp}}(P_{\perp})$ agrees with the banked campaign table to between 8.8×10^{-9} and 1.8×10^{-6} relative at every size, which is the rounding of the six-figure constants compared against, and ice-rule and winding violations were zero throughout. Write $R_h(L) = \tau_{\text{exp}}(h, q_{\min}) / \tau_{\text{exp}}(P_{\perp}, q_{\min})$.

Table 3: Cubic ladder, 1024 replicas (512 at $L = 16$), 6,000 samples per replica, jackknifed over multi-spin-coding blocks.

L	2	3	4	6	8	12	16
R_h	0.6813	0.5417	0.5131	0.5048	0.5026	0.5010	0.4988
$\pm$	0.0051	0.0023	0.0018	0.0017	0.0019	0.0015	0.0025
$ P_{\perp} ^2 / P_{\perp}$	0.4608	0.4687	0.4847	0.4928	0.4958	0.4992	0.4972
$\tau_{\text{exp}}(P_{\perp})$ (sw)	1.8031	3.4722	5.8044	12.428	21.813	48.424	85.929

R_h falls monotonically to one half, approaching from above and crossing to $0.4988 \pm$

0.0025 at $L = 16$, while the independent control $|P_\perp|^2/P_\perp$ approaches the same value from below, the two meeting by $L = 8$. All four helicity families pass the resolution gate at every size, with fit r^2 between 0.9891 at $L = 2$ and 1.0000 for $L \geq 6$, so no helicity number here is a fit to noise. At the Gaussian fixed point the photons are odd under arrow reversal, so the even sector is the sector of even photon number, whose lowest state is two photons at $\pm q_{\min}$ relaxing at half the odd time. The helicity is a two-photon composite and lands exactly there. It had no obligation to, since it carries a different representation of the cubic point group from $|P_\perp|^2$, and the two agreeing from opposite sides is better evidence than either alone. In cubic geometry the odd polarisation remains the slowest member of the panel at every size.

7 Why the thin cell is different

Thinness alone is not the cause. Across the cells the enumerator can reach, $\Delta_{\text{even}}/\Delta_{\text{odd}}$ sorts by cross-section rather than by volume.

Table 4: $\Delta_{\text{even}}/\Delta_{\text{odd}}$ per attempt on the giant component of each enumerable cell. The $(1, 1, c)$ cells are quasi-one-dimensional and were excluded from anchoring anything, and at $(1, 1, 4)$ the giant component is not the component the production chain lives in.

cell	spins	giant	quasi-1D	$1/\Delta$ even	$1/\Delta$ odd	ratio
(1, 1, 2)	32	160	yes	35.67	132.47	3.71429
(1, 1, 3)	48	3,360	yes	81.48	310.56	3.81136
(1, 2, 2)	64	221,424	no	389.95	242.57	0.62206
(1, 1, 4)	64	43,008	yes	161.85	628.03	3.88035

The three quasi-one-dimensional cells sit between 3.7 and 3.9 and rise slowly with length. $(1, 2, 2)$ is the only reachable cell with two directions of extent greater than one, and the only one where the even sector is slowest. A helicity needs two independent transverse directions to have a long wavelength, and the measurements say so directly: the helicity about the thin axis has overlap 0.03464 with v against 0.75859 for the combination

on the two thick axes. Momentum is the other half of the geometry. With $b = c = 2$ the smallest nonzero wavevector along a transverse axis is already the zone boundary, where $e^{iq_{\min}a} = -1$, and that is exactly where the character analysis puts the fourfold odd multiplet while the helicity sits at zero momentum. A cell two units across has no interior to its Brillouin zone, so the ordering of the two sectors is settled by a two-point comparison a larger cell would not make. Adding a third thick direction restores the Gaussian counting, and the cubic ladder shows that happening.

8 What remains open

The natural next cell is $(1, 2, 3)$, which would test whether the helicity stays slowest when one transverse direction is lengthened. Its $\mathbf{W} = 0$ count is known exactly, 196,465,480 states from a frontier dynamic programme in 14.9 seconds. The diagonalisation is what does not fit, at 31 to 243 hours of wall time, the binding constraints being the binary-search index used to build the adjacency and the unaccelerated Lanczos iteration count. That cell would still be one unit thick, so it would sharpen the thin-cell statement rather than test cubic geometry.

The cubic result is a measurement supported by a field-theoretic argument, and a panel bounds only what it measures. An eighth family finding nothing slower than the polarisation at seven sizes narrows the space of hidden even modes without emptying it. Nothing here is certified. The spectra are floating-point computations with residual checks and cross-checks against independently banked values, the symmetry statements are numerical characters agreeing with $\det M_g$ to 2.7×10^{-14} , and the cubic result is Monte Carlo with jackknife errors, which is a different epistemic status from the project's SAT-certified and proof-assistant results.

9 Related work

Magnetic helicity entered fluid mechanics as a topological invariant. Woltjer proved the invariance of $\int \mathbf{A} \cdot \mathbf{B}$ for a perfectly conducting fluid and its role in characterising force-free fields (Woltjer 1958); Moffatt showed that the corresponding integral for a vortex field measures the knottedness and linkage of the field lines (Moffatt 1969); Berger and Field

made the topological reading precise for a magnetic field in a bounded volume (Berger and Field 1984). The pseudoscalar character that does the identifying work here is the property that makes helicity a linking number, since a linkage changes sign under reflection.

The emergent electrodynamics of spin ice is why a helicity exists in this model at all. The ice rule is a lattice divergence-free condition on the coarse-grained field, which puts the classical model in a Coulomb phase with dipolar correlations and pinch points in the structure factor (Isakov et al. 2004; Henley 2010), and the excitations above it are deconfined monopoles (Castelnovo, Moessner, and Sondhi 2008, 2012). At the Rokhsar–Kivelson point the quantum model has an emergent photon (Hermele, Fisher, and Balents 2004), and the classical-to-quantum correspondence (Henley 2004) makes the chain’s spectrum that Hamiltonian’s spectrum. I have found no previous use of the helicity as an observable in spin-ice Monte Carlo, classical or quantum: a literature search on the emergent gauge field returns work on the transverse projector, on monopoles and on photon signatures, and nothing on pseudoscalar bilinears. That is a negative result from one search rather than a claim of priority.

The general point that an observable’s autocorrelation time is not the chain’s is old (Sokal 1997), and the practical version is familiar in lattice gauge theory, where the topological charge is a slow mode that couples weakly to most observables and so leaves their error estimates too small (Schaefer, Sommer, and Virotta 2011). The pyrochlore case is sharper than weak coupling, because the overlap is exactly zero and two symmetries in succession enforce it: arrow reversal removes every odd observable, and the improper operations of the point group remove every even scalar. Where that matters, the alternative to guessing a better observable is to compute the subdominant eigenvalue of the stochastic matrix itself, the route Nightingale and Blöte took for the two-dimensional Ising model (Nightingale and Blöte 1996), and what the diagonalisation here does at the one cell where it is affordable.

10 Appendix: artefacts

All computation was CPU-only on a single GB10 machine, so no HPC service was used and no acknowledgement is owed. `icemix/parity.py` supplies the parity blocks, the derived automorphism group, the induced action on states and the monomial projection; `icemix/observables.py` defines $P_j(\mathbf{q})$; `icemix/exact_ac.py` computes exact auto-

correlations and overlaps; `icemix/panel.py` carries the panel, the helicity families and `helicity_ratio.experiments/05_thin_even/results/results.json` is the source of every $(1, 2, 2)$ number here, from a run of 886.9 s at a peak resident set of 4.45 GiB, and `experiments/06_helicity_cubic/results/results.json` the source of every cubic number, from a run of 3,413.5 s at 8.21 GiB. `tests/test_stage4d.py` adds 26 tests and `tests/test_stage4f.py` 17, covering the parity and symmetry machinery from definitions, the panel change and its bit-identity against the pre-change module, and the recorded readings. `docs/reports/2026-09-04-stage4d.md` and `docs/reports/2026-09-04-stage4f.md` carry the full working record.

References

- Berger, Mitchell A., and George B. Field (1984), ‘The Topological Properties of Magnetic Helicity’, *Journal of Fluid Mechanics*, 147: 133–48, <https://doi.org/10.1017/S0022112084002019>.
- Bramwell, Steven T., and Michel J. P. Gingras (2001), ‘Spin Ice State in Frustrated Magnetic Pyrochlore Materials’, *Science*, 294: 1495–501.
- Castelnovo, Claudio, Roderich Moessner, and Shivaji L. Sondhi (2008), ‘Magnetic Monopoles in Spin Ice’, *Nature*, 451: 42–5.
- Castelnovo, Claudio, Roderich Moessner, and Shivaji L. Sondhi (2012), ‘Spin Ice, Fractionalization, and Topological Order’, *Annual Review of Condensed Matter Physics*, 3/1: 35–55, <https://doi.org/10.1146/annurev-conmatphys-020911-125058>.
- Henley, Christopher L. (2004), ‘From Classical to Quantum Dynamics at Rokhsar–Kivelson Points’, *Journal of Physics: Condensed Matter*, 16: S891–8, <https://arxiv.org/abs/cond-mat/0311345>.
- Henley, Christopher L. (2010), ‘The “Coulomb Phase” in Frustrated Systems’, *Annual Review of Condensed Matter Physics*, 1: 179–210, <https://doi.org/10.1146/annurev-conmatphys->

070909-104138.

Hermele, Michael, Matthew P. A. Fisher, and Leon Balents (2004), 'Pyrochlore Photons: The $U(1)$ Spin Liquid in a $S = 1/2$ Three-Dimensional Frustrated Magnet', *Physical Review B*, 69: 064404, <https://arxiv.org/abs/cond-mat/0305401>.

Isakov, S. V., et al. (2004), 'Dipolar Spin Correlations in Classical Pyrochlore Magnets', *Physical Review Letters*, 93: 167204, <https://doi.org/10.1103/PhysRevLett.93.167204>.

Moffatt, H. K. (1969), 'The Degree of Knottedness of Tangled Vortex Lines', *Journal of Fluid Mechanics*, 35/1: 117–29, <https://doi.org/10.1017/S0022112069000991>.

Nightingale, M. P., and H. W. J. Blöte (1996), 'Dynamic Exponent of the Two-Dimensional Ising Model and Monte Carlo Computation of the Subdominant Eigenvalue of the Stochastic Matrix', *Physical Review Letters*, 76: 4548–51, <https://doi.org/10.1103/PhysRevLett.76.4548>.

Rokhsar, Daniel S., and Steven A. Kivelson (1988), 'Superconductivity and the Quantum Hard-Core Dimer Gas', *Physical Review Letters*, 61: 2376–9.

Schaefer, Stefan, Rainer Sommer, and Francesco Virotta (2011), 'Critical Slowing down and Error Analysis in Lattice QCD Simulations', *Nuclear Physics B*, 845/1: 93–119, <https://doi.org/10.1016/j.nuclphysb.2010.11.020>.

Sokal, Alan (1997), 'Monte Carlo Methods in Statistical Mechanics: Foundations and New Algorithms', in *Functional Integration: Basics and Applications*, NATO ASI Series (Boston, MA), 131–92, https://doi.org/10.1007/978-1-4899-0319-8_6.

Woltjer, L. (1958), 'A Theorem on Force-Free Magnetic Fields', *Proceedings of the National Academy of Sciences*, 44/6: 489–91, <https://doi.org/10.1073/pnas.44.6.489>.